Brian Thomas Cook

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Education

University of North Carolina at Chapel Hill *2028 (expected)*
Astrophysics PhD

Leiden University *2020*
Astronomy MSc, specialization in Astronomy Research

Georgia Institute of Technology *2018*
Physics MS

University of Michigan *2017*
Physics BS with Honors, Astronomy & Astrophysics BS

Publications

Total citations (via Google Scholar): 27, h-index: 4.

First-author papers

3.) **B.T. Cook**, K. Tep, C.L. Rodriguez, *et al.*: "Modeling Globular Cluster Stellar Streams with a Basis-Expansion N-body Code", Astrophysical Journal 1010, 155 (2026).

2.) **B.T. Cook**, D.F. Woods, J.D. Ruprecht, *et al.*: "Tracing Milky Way substructure with an RR Lyrae hierarchical clustering forest", Monthly Notices of the Royal Astronomical Society 513, 2 (2022).

1.) **B.T. Cook**, J.J. Tobin, M.F. Skrutskie, and M.J. Nelson: "Time variability in the bipolar scattered light nebula of L1527 IRS: a possible warped inner disk", Astronomy & Astrophysics 626, A51 (2019).

Second-Nth author papers

2.) N. Panithanpaisal, R.E. Sanderson, C.L. Rodriguez (+6 co-authors and **B.T. Cook**) "Breaking Down the CosmoGEMS: Toward Modeling and Understanding Globular Cluster Stellar Streams in a Fully Cosmological Context", Astrophysical Journal 997, 182 (2026).

1.) K. Tep, **B.T. Cook**, C.L. Rodriguez, *et al.*: "KRIOS: A new basis-expansion N-body code for collisional stellar dynamics", Astrophysical Journal 993, 180 (2025).

Presentations

Talks

57th DDA Meeting, 2026.
Cosmo-Astro Seminar, U. of Michigan, 2026.
Stars Meeting, CIERA, 2025.
Theory Meeting, CIERA, 2025.
Coffee Hour, Carnegie Observatories, 2025.
56th DDA Meeting, 2025.
North Carolina Space Symposium, 2025.
Master's Colloquium, Leiden Observatory, 2020.
Cosmic Coffee, Center for Relativistic Astrophysics, Georgia Tech, 2019.
Group 97 Technical Seminar, Space Systems and Technology Division, MITLL, 2019.

Poster Presentations

Dwarf Galaxies, Star Clusters, and Streams in the LSST Era, KICP, 2024.

229th AAS Meeting, 2017.

Relevant Work Experience

Graduate Research Assistant, UNC-Chapel Hill Department of Physics and Astronomy: Lead developer and project manager for KRIOS, an N-body code written in C++ designed to simulate large-N self-gravitating systems. (*08/2023 - present.*)

Data Scientist, Entertainment Data Oracle, Inc.: Made several improvements to EDO's automated content recognition software. Created an unsupervised clustering routine to deliver reports on small creative differences. Created a model trained on thousands of hours of content capable of programmatically identifying commercial and programming blocks. (*09/2021 - 08/2023.*)

Graduate Teaching Assistant, Georgia Tech School of Physics: Taught introductory physics courses designed for undergraduate engineering and biological science majors. These assignments involved helping students with analytic problems, collecting data in a variety of contexts, and writing VPython source code. (*08/2017 - 08/2018, 08/2020 - 05/2021.*)

Summer Research Program Intern, Applied Space Systems Group, MIT Lincoln Laboratory: Used hierarchical clustering to discern structures containing RR Lyrae variable stars in the Milky Way's stellar halo. Contributed a random forest classification element to an optical image data analysis pipeline as part of an existing collaboration with scientists at the Jet Propulsion Laboratory and Caltech. (*06/2019 - 08/2019.*)

Honors and Awards

Graduate Research Fellowship, North Carolina Space Grant.

President's Fellowship, Georgia Tech.

Sigma Pi Sigma, U. of Michigan chapter.

Service

Referee, Astrophysical Journal